159102 workshop – Week 5

1. Write down any changes to registers, the memory locations and the program counter when the following MASSEY machine programs are run:

Program (a):

```
130F
8302
4383
E000
```

Program (b):

```
11E0
9108
4103
D1DD
```

What is interesting about this program?

2. Write the following program as assembler instructions. This will make it easier to read.

```
1032
11FF
2071
9108
B117
C201
1301
6423
6504
E000
```

3. Use the assembler code in (2) above to identify what is happening in the program.

a) What value is stored in register 5 and why?

b) Can you write the program to achieve the same result in a different way and thereby use less machine code instructions?

4. Write a MASSEY machine code program that is the equivalent of the following C++ code:

```
i = 9;
while (i >= 0) {
  i--;
}
```